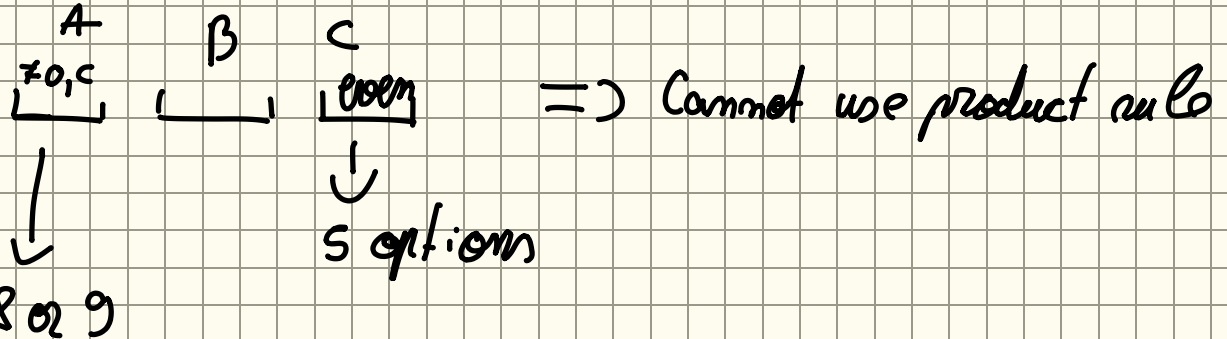
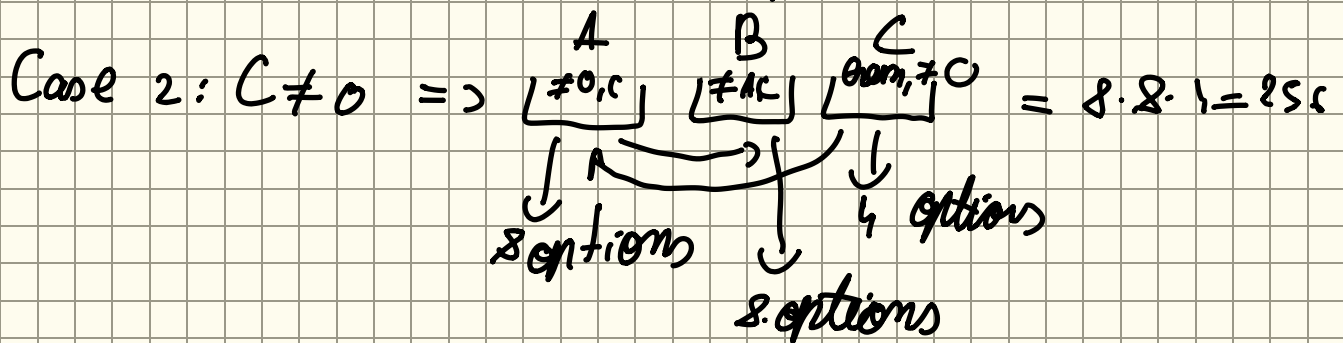
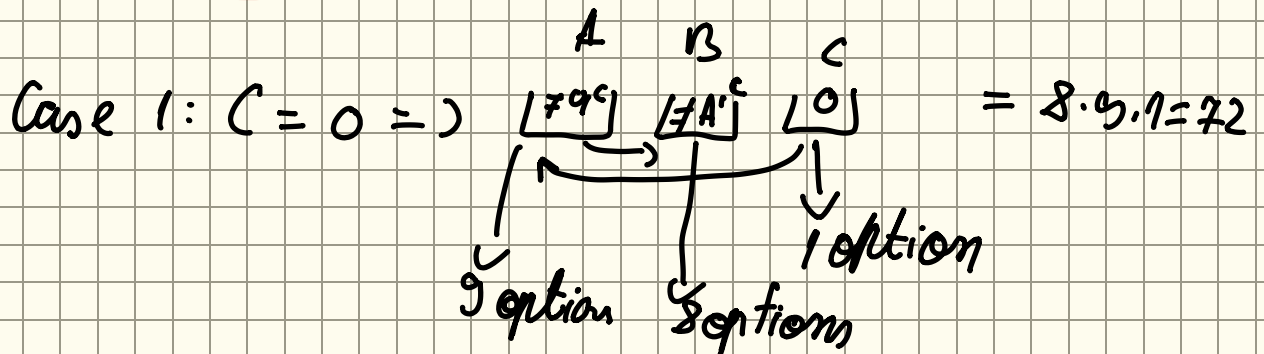


Enumerative combinatorics

ex. How many even 3-digit numbers have all their digits distinct?



↳ The sum rule:



$$\Rightarrow 72 + 256 = 328$$

↳ The complement rule (the difference rule):

$$9 \cdot 9 \cdot 8 = 648 \text{ numbers distinct digits}$$

$$8 \cdot 8 \cdot 5 = 320 \text{ odd numbers distinct digits}$$

$$\Rightarrow 648 - 320 = 328 \text{ even } \underline{\hspace{1cm}} // \underline{\hspace{1cm}}$$

ex. How many 3-digit numbers that contain the number 7 at least one

$$\begin{array}{l} \text{all: } 9 \cdot 10 \cdot 10 = 900 \\ \text{Without 7: } 8 \cdot 9 \cdot 9 = 648 \end{array} \quad \Rightarrow \quad \text{With 7: } 900 - 648 = 252$$

k-permutations and permutations without repetition: